MEDRECORDS

Product Requirements Document

Nigeria's Unified Patient-Owned Medical Record Platform

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1. Executive Summary

MedRecords is a patient-owned, consent-driven Electronic Health Record (EHR) and Personal Health Record (PHR) platform for Nigeria. It gives every Nigerian a single portable medical identity that follows them across hospitals, clinics, pharmacies, dental practices, optical centers, laboratories, and specialists — replacing today's fragmented paper system with a structured, queryable, exportable record.

The platform is delivered as a multi-tenant SaaS with mobile-first patient apps, web and desktop apps for providers, and HL7 FHIR APIs for institutional integration. It is designed for Nigerian operating conditions: intermittent connectivity, low-feature devices in rural settings, NIN-based national identity, NDPA-regulated data handling, and a provider market dominated by private clinics and pharmacies operating on thin margins.

This PRD defines what we are building in the first 18 months. It explicitly addresses the gaps in the original concept document: a real identity strategy (NIN-anchored), a granular and break-glass-capable consent model, FHIR as a Day-1 foundation rather than a future ambition, offline-first design, and an adoption strategy that confronts the central commercial question — who pays, in what order.

2. Problem Statement & Market Context

2.1 The Status Quo

- Medical records in Nigeria are predominantly paper-based, stored within the four walls of the institution that created them. The patient owns nothing portable.
- There is no canonical patient identifier in clinical use. Hospitals issue their own MRNs (Medical Record Numbers); the National Identification Number (NIN) is rarely linked to clinical records.
- Continuity of care across providers is essentially manual. Patients carry physical folders, X-ray envelopes, and prescription stubs between facilities, or simply re-explain their history each time.
- Pharmacies dispense without visibility into other prescriptions, creating drug interaction risk. Lab results are paper printouts handed back to the patient.
- Embassies, visa offices, employers, and overseas hospitals routinely require structured medical history that Nigerians cannot easily produce.
- Existing hospital management systems (HMS) are siloed, vendor-locked, and rarely interoperable. None aggregate across institutions at the patient level.

2.2 Why Now

- NIN coverage has crossed 100 million registered Nigerians, making national identity infrastructure usable as an anchor.
- Smartphone penetration is sufficient in urban centers to support a mobile-first patient experience, with a meaningful USSD/feature-phone fallback layer required for rural reach.
- The Nigeria Data Protection Act 2023 (NDPA) and the National Health Act 2014 together create a regulatory framework that legitimizes — and to some extent mandates — secure digital health records.
- Global FHIR adoption gives us a tested interoperability standard rather than forcing us to invent one.
- Health insurance penetration is growing (NHIA, state schemes, private HMOs), creating institutional buyers who need claims-grade clinical data.

2.3 Stakeholders

MedRecords sits at the intersection of five constituencies, each with different incentives and friction points:

Stakeholder	What they need from MedRecords
Patients	Want continuity, portability, control. Will not pay much directly.
Healthcare providers	Want speed and accuracy. Will resist any tool that slows clinical workflow.
Payers (HMOs, NHIA, state schemes)	Want claims-grade clinical data to reduce fraud and improve risk pricing. Have budget.
Regulators (NDPC, FMOH, NHIA)	Want compliance, auditability, and population health insight. Have authority.
Government / national programs	Want disease surveillance, vaccination tracking, and a usable digital health backbone. Have scale.

3. Goals, Non-Goals & Guiding Principles

3.1 Goals (First 18 Months)

- G1 — Launch a working patient app (iOS, Android, USSD fallback) with a verified MedRecords ID anchored to NIN.
- G2 — Onboard at least 50 active provider sites across hospitals, pharmacies, and specialty clinics in Lagos and one secondary city.
- G3 — Ingest at least 250,000 structured clinical records (encounters, prescriptions, lab results, dispenses) within FHIR-compatible schemas.
- G4 — Demonstrate continuity-of-care: at least 30% of active patients have records from two or more distinct providers.
- G5 — Achieve NDPA registration as a data controller and complete an initial third-party security audit.
- G6 — Secure at least one institutional payer pilot (HMO or state insurance scheme).

3.2 Non-Goals (Explicit Exclusions for v1)

- Not a Hospital Management System (HMS). We do not bill, manage beds, schedule shifts, or replace existing hospital ERPs. We integrate with them.
- Not a telemedicine platform. Consultation video, chat, or appointment booking are out of scope for v1.
- Not a diagnostic AI product. AI assistance is a v3+ consideration once we have data and trust.
- Not an insurance claims engine. We provide the data layer that claims systems can consume; we do not adjudicate claims.
- Not a pharmacy POS or inventory system. We integrate with dispensing workflows but do not replace pharmacy software.

3.3 Guiding Principles

- Patient-owned by default. The patient is the data controller. Providers are processors operating under consent.
- Standards-first. FHIR R4 is the canonical internal data model from Day 1, not a future migration.

- Offline-first. The patient app and provider apps must function during network loss and reconcile when reconnected.
- Workflow-respecting. We do not add steps to a clinician's day. If integration takes longer than the current paper workflow, we have failed.
- Auditable always. Every read and every write is logged. Patients can see who looked at their data and when.

4. Target Users & Personas

Adaeze, 34 — Working professional in Lagos, mother of two

Manages her own healthcare across a private hospital, a pediatrician for her kids, an ophthalmologist, and a pharmacy near her office. Has gestational diabetes history from her last pregnancy.

Jobs to be done: View and share her own and her children's records, refill prescriptions safely, share history with a new doctor when changing providers, export records for a US visa application.

Pain points: Loses old prescriptions and lab results, repeats tests, has to retell history every visit, no proof of vaccination dates for her children's school.

Dr. Tunde, 41 — General practitioner at a 40-bed private hospital in Ibadan

Sees 25–40 patients per day. Has a half-functional desktop HMS for billing but writes clinical notes by hand.

Jobs to be done: Quickly see prior conditions, current medications, and allergies for a new patient; document encounter; print prescription.

Pain points: First-visit patients give incomplete histories; pharmacy dispenses without coordination; referrals to specialists disappear into a black hole.

Mrs. Okafor — Pharmacist and owner of a 3-branch retail pharmacy chain

Dispenses 200–400 scripts per day across branches. Currently uses paper prescriptions and a basic POS.

Jobs to be done: Verify a prescription is real and not duplicated, record what was dispensed, flag dangerous interactions, build customer loyalty through refill reminders.

Pain points: Cannot tell if a prescription has been filled elsewhere; controlled substances are a compliance risk; loses repeat business to whichever pharmacy is closest.

Ifeoma, NHIA Operations Lead — Manages a 200,000-life enrolment scheme

Needs structured encounter and prescription data from accredited providers to validate claims and detect fraud.

Jobs to be done: Pay providers faster with less manual review; identify outlier providers; report disease burden upward.

Pain points: Today she relies on paper claims and self-reported diagnosis codes; fraud is estimated to consume a significant share of claims spend.

5. Product Scope

5.1 In Scope (v1, first 18 months)

- Patient mobile app (iOS, Android) with USSD fallback for non-smartphone users
- Provider web app (clinician workstation) with offline-capable PWA mode

- Pharmacy module (prescription verification + dispense recording)
- Specialist module with role-based record visibility (dental, optical, chiropractic, lab)
- Identity verification via NIN + biometric/photo verification
- Consent management (granular, time-bound, break-glass)
- FHIR R4 REST API for institutional integration
- Audit log with patient-facing access history
- Document export (PDF) for embassies, employers, foreign hospitals
- Admin console for provider management, audit, billing

5.2 Out of Scope (v1)

- Telemedicine / video consultations
- Insurance claims adjudication
- Pharmacy inventory and POS
- AI diagnosis or clinical decision support beyond simple drug-interaction flags
- Wearable / IoT device integration
- Patient-to-patient messaging or social features

6. Functional Requirements

Requirements are grouped by module. Each requirement has an ID, priority (P0 = launch-blocking, P1 = v1 must-have, P2 = v1.x), and acceptance criteria.

6.1 Patient App

ID	Pri	Requirement
FR-PA-001	P0	Account creation via NIN verification + phone OTP. Biometric/photo match against NIMC.
FR-PA-002	P0	View complete medical history grouped by record category (see §9).
FR-PA-003	P0	Grant, modify, and revoke consent to specific providers. Consent options: full / category-limited / time-limited.
FR-PA-004	P0	View audit log: who accessed my record, when, what they viewed, under what consent.
FR-PA-005	P0	Generate a one-time access code (QR + 6-digit number) for in-person provider access.
FR-PA-006	P1	Manage dependents (children under 16, dependent adults with proof of guardianship).
FR-PA-007	P1	Export medical history as signed PDF with QR verification link.
FR-PA-008	P1	Prescription refill reminders and medication adherence tracking.
FR-PA-009	P1	Offline view of cached records (last 90 days).

ID	Pri	Requirement
FR-PA-010	P2	Allergy and chronic condition flags surfaced to any provider on consent grant.
FR-PA-011	P2	USSD shortcode for record summary and consent grant on feature phones.

6.2 Provider (Clinician) App

ID	Pri	Requirement
FR-CL-001	P0	Verify patient identity via QR/code scan or NIN+OTP, with patient consent capture.
FR-CL-002	P0	View patient record filtered to the consented scope and the clinician's specialty.
FR-CL-003	P0	Create structured encounter: chief complaint, diagnosis (ICD-10), vitals, plan, free-text notes.
FR-CL-004	P0	Issue electronic prescriptions with drug name, strength, dose, route, frequency, duration.
FR-CL-005	P0	Order labs and attach results when received (manual upload or lab integration).
FR-CL-006	P0	Offline mode: queue encounters and prescriptions; sync on reconnect; conflict resolution UI.
FR-CL-007	P1	Break-glass emergency access with mandatory justification and immediate patient notification.
FR-CL-008	P1	Refer to specialist with structured referral letter pre-populated from record.
FR-CL-009	P1	Template library for common encounter types per specialty.
FR-CL-010	P2	Voice-to-text dictation for clinical notes (English + major Nigerian languages).

6.3 Pharmacy Module

ID	Pri	Requirement
FR-PH-001	P0	Verify prescription by scanning QR or entering prescription ID + patient consent.
FR-PH-002	P0	Display only: prescription details, known allergies, active prescriptions from other providers.
FR-PH-003	P0	Record dispense event (full or partial, with quantity).
FR-PH-004	P0	Flag dangerous interactions with active prescriptions before dispense.
FR-PH-005	P1	Flag duplicate dispense attempts (prescription already filled elsewhere).

ID	Pri	Requirement
FR-PH-006	P1	Substitution suggestions (generic equivalents) with prescriber notification.
FR-PH-007	P2	Controlled substance register synced to NAFDAC reporting requirements.

6.4 Specialist Module (Dental, Optical, Chiropractic, Lab)

ID	Pri	Requirement
FR-SP-001	P0	Specialty-scoped view of patient record (e.g., dentist sees dental + allergies + relevant meds).
FR-SP-002	P0	Upload imaging and procedure records to the appropriate record category.
FR-SP-003	P1	Specialty-specific encounter templates (dental chart, refraction, spinal exam, etc.).
FR-SP-004	P1	Cross-specialty alert: flag findings that other providers should know about.

6.5 Integration & Admin

ID	Pri	Requirement
FR-IN-001	P0	FHIR R4 REST API: Patient, Encounter, Condition, Observation, MedicationRequest, MedicationDispense, AllergyIntolerance, Immunization, DiagnosticReport, DocumentReference.
FR-IN-002	P0	HMS integration adapters for at least two market-leading Nigerian HMS vendors.
FR-IN-003	P0	Provider admin console: user management, role assignment, billing, audit export.
FR-IN-004	P1	Payer API: scoped access for HMOs/NHIA to validate enrolled members' encounters.
FR-IN-005	P1	Webhook subscriptions for record events (consented downstream systems only).
FR-IN-006	P2	Bulk export for population health research (de-identified, with IRB approval workflow).

7. Identity Model

Detailed in the companion Consent & Identity Model document. Summary follows.

7.1 Identity Anchor: NIN

Every MedRecords account is anchored to a verified National Identification Number. NIN provides national uniqueness, government-backed identity assurance, and a path to interoperability with other government health systems (e.g., NHIA enrolment).

Verification combines: (a) NIN lookup via NIMC API, (b) phone OTP to the registered MSISDN, (c) facial photo match against NIMC's recorded photo (where consented and where the API path permits).

7.2 Identity Edge Cases

- Patients without NIN — degraded mode ("MedRecords Provisional ID") with limited features and a 6-month deadline to verify, after which the account is frozen but data preserved.
- Children under 16 — no independent account. Linked as dependents under a verified guardian's account, with automatic promotion at age 16.
- Dependent adults — guardian linkage requires uploaded proof (power of attorney, court order, or medical incapacity certificate).
- Deceased patients — record is preserved and frozen. Executor access requires probate documentation, processed manually for v1.
- Visitors / foreigners — passport-based provisional account, restricted to outpatient and emergency scenarios.

7.3 Provider Identity

Providers (clinicians, pharmacists, dentists) are verified against their respective professional registries: MDCN (medical doctors), PCN (pharmacists), MDCN-dental wing, ODORBN (optometry), etc. Provider accounts are tied to both the practitioner and one or more registered facilities.

8. Consent Model

Detailed in the companion Consent & Identity Model document. Summary follows.

8.1 Consent Types

Type	Description
Full	Provider can see all record categories. Used for primary care relationships.
Category-limited	Provider sees only relevant categories (e.g., dentist sees dental + allergies + active meds).
Time-limited	Access expires automatically after a defined window (default 24h for visits, 7d for ongoing episodes).
Single-encounter	One-time read access tied to a specific visit; auto-expires when the encounter is closed.
Break-glass (emergency)	Provider can self-authorize emergency access with mandatory justification. Patient is notified immediately. Full audit and post-hoc review.

8.2 Consent Lifecycle

- Grant — patient initiates via app, QR scan, or in-clinic signed form digitized by provider (with patient OTP confirmation).

- Active — provider may read/write within the consent scope. Every action is logged.
- Modify — patient may narrow scope or shorten duration at any time. Provider receives notification.
- Revoke — patient may revoke at any time. Past records the provider authored remain authored by them but are no longer readable by them.
- Expire — time-limited consents auto-expire. Provider receives 24-hour warning.
- Audit — patient sees a chronological feed of every consent grant, access event, and revocation.

8.3 Break-Glass Rules

- Available only to verified providers at registered emergency-capable facilities.
- Requires the provider to enter a justification (free text + dropdown reason code).
- Patient receives immediate SMS and push notification with provider name, facility, and timestamp.
- Auto-expires after 24 hours unless converted to standard consent.
- All break-glass events are reviewed by MedRecords compliance team within 7 days, with sampled review of justifications.
- Repeated break-glass by the same provider without subsequent patient consent triggers an automatic facility audit.

9. Data Model & Record Categories

All clinical data is stored using FHIR R4 resources as the canonical model. The platform exposes a category abstraction ("tabs") for patient and provider UIs, mapped from FHIR resources.

Category (UI tab)	Canonical FHIR resources
General Medical History	Patient, Condition, FamilyMemberHistory
Encounters & Visits	Encounter, ClinicalImpression, CarePlan
Surgeries & Procedures	Procedure
Hospitalizations	Encounter (class=IMP), EpisodeOfCare
Prescriptions	MedicationRequest
Pharmacy Dispenses	MedicationDispense
Dental Records	Procedure, Observation, DocumentReference (charts/X-rays)
Eye Care	Observation (refraction), VisionPrescription
Lab Results	DiagnosticReport, Observation, Specimen
Vaccinations	Immunization
Allergies & Conditions	AllergyIntolerance, Condition
Referrals	ServiceRequest, DocumentReference
Documents & Images	DocumentReference, ImagingStudy, Media

ICD-10 is used for diagnosis coding; SNOMED CT for clinical terms where available; NAFDAC drug codes mapped to RxNorm for medication; LOINC for lab observations.

10. Non-Functional Requirements

Attribute	Target
Availability	99.5% monthly uptime in v1, 99.9% by month 18
Latency	Provider record load < 2s (P95) on broadband; < 5s on 3G; offline read instant from cache
Scale (v1 target)	1M patient accounts, 5K active providers, 50M records, 500K daily API calls
Data residency	All PII and clinical data stored within Nigeria (per NDPA Section 41). Multi-AZ within the country.
Backup & DR	RPO 1 hour, RTO 4 hours. Geographically separated in-country backups.
Mobile size	Patient APK under 30MB. Works on Android 8+.
Accessibility	WCAG 2.1 AA for web; large-text and high-contrast modes on mobile.
Localization	English at launch; Hausa, Yoruba, Igbo by month 12 for patient app.

11. Security, Privacy & Compliance

11.1 Regulatory Framework

- Nigeria Data Protection Act 2023 (NDPA) — controller registration, DPO appointment, breach notification within 72 hours, data subject rights.
- National Health Act 2014 — patient rights, confidentiality obligations, record retention.
- Medical and Dental Council of Nigeria (MDCN) Code of Ethics — clinician duties around records.
- Pharmacists Council of Nigeria (PCN) regulations — dispensing records.
- International alignment: HIPAA-equivalent controls, ISO 27001 target certification, SOC 2 Type II by month 24.

11.2 Controls

- Encryption: AES-256 at rest (database, blob storage, backups); TLS 1.3 in transit; field-level encryption for the highest-sensitivity fields (e.g., HIV status, mental health diagnoses) with separate key custody.
- Authentication: OAuth2/OIDC, mandatory MFA for all provider accounts, biometric/passkey support for patient app.
- Authorization: RBAC for provider roles; ABAC for specialty-scoped record visibility; per-resource consent enforcement.
- Audit: immutable append-only audit log for every read and write, accessible to the patient and to compliance.
- Penetration testing: independent annual pentest; quarterly automated vulnerability scans; bug bounty program from month 12.

- Incident response: documented runbook; on-call rotation; mandatory tabletop exercise quarterly.
- Key management: cloud KMS with hardware-backed root keys; quarterly key rotation; split-knowledge admin procedures.

12. Interoperability

FHIR R4 is the canonical internal data model. The public API is HL7 FHIR REST. We publish an Implementation Guide (IG) profiling the Nigeria-specific extensions: NIN identifier, NAFDAC drug codes, MDCN/PCN practitioner identifiers, Nigerian address formats.

HMS integration is delivered through pre-built connectors for the top Nigerian HMS vendors (final list determined during partner discovery, but the technical pattern is identical: a thin adapter that translates the HMS's native model into our FHIR profile).

We will join the global FHIR community and align with WHO SMART Guidelines where applicable (vaccination, antenatal care, HIV).

13. Offline-First & Connectivity Strategy

Connectivity in Nigeria varies from gigabit fiber in Victoria Island to no signal in rural LGAs. The platform must function across that range.

13.1 Patient App

- Last 90 days of records cached locally and available offline.
- Consent grants for in-person visits work offline using cryptographically signed local tokens, validated on next sync.
- Export PDF works offline from cached records.

13.2 Provider App

- Full encounter authoring works offline. Encounters queue locally and sync when connectivity returns.
- Patient records are pre-fetched on consent grant, even if connectivity drops mid-visit.
- Conflict resolution UI for the rare case of concurrent edits across two offline providers.

13.3 USSD Fallback

- Short code (e.g., *xxx#) provides: account creation, consent grant via session code, last-3 records summary, prescription verification.
- Useful for patients on feature phones and for emergencies when the app is unavailable.

14. Provider Onboarding & Adoption

The single biggest risk to MedRecords is not technical — it is provider adoption. The platform fails if clinicians don't input data. Our adoption strategy is detailed in the companion Business Model & GTM document. Summary follows.

14.1 Wedge: Pharmacies First

Pharmacies have the simplest workflow integration (verify prescription → dispense → record), the clearest commercial incentive (loyalty, repeat business, regulatory compliance on controlled substances), and the highest transaction frequency per location. They generate the dispensing data that gives the platform initial breadth.

14.2 Second Wave: GP Clinics in Concentrated Areas

After pharmacy density in a target area (e.g., a Lagos LGA) crosses a threshold, we approach the GP clinics in that area with a value proposition that depends on the pharmacy data already in the system: when you prescribe, you know what was dispensed; when a new patient walks in, you can see what they were prescribed elsewhere.

14.3 Third Wave: Specialists, Labs, and Institutional Partners

Once the GP layer is producing structured encounter data, specialty providers (dentists, optometrists, labs) are easier to onboard because the referral and result flow already exists. Institutional payers (HMOs, NHIA) become the long-term revenue base.

15. Business Model Summary

Full detail in the companion Business Model & GTM document. Summary:

Customer segment	Pricing approach
Patients	Free for individuals. Premium tier for families with extended export and 24/7 support.
Pharmacies	Tiered SaaS per dispensing point; free entry tier to drive adoption.
Clinics & Hospitals	Per-provider SaaS with volume discounts; integration fees for HMS adapters.
HMOs / Payers	Per-member-per-month data access for enrolled lives.
API / Integration partners	Usage-based for high-volume integrators; revenue share for HMS vendors.

16. Success Metrics & KPIs

16.1 North-Star Metric

Cross-provider continuity rate: the percentage of active patient accounts that have structured records from two or more distinct providers within the trailing 12 months. This single number captures whether MedRecords is actually doing what it claims — unifying the record.

16.2 Supporting Metrics

- Verified patient accounts (NIN-anchored)
- Active providers (logged in and authored a record in trailing 30 days)
- Records ingested per week, by category
- Consent grants per week; consent revocations per week (a healthy ratio matters)

- Provider time-to-first-record (onboarded → first authored encounter)
- API uptime and P95 latency
- NPS — patient and provider, tracked separately
- Break-glass events per 1,000 active patients (should be very low)
- Audit log query rate (patients checking who accessed their record — healthy engagement signal)

17. Risks & Mitigations

Risk	Mitigation
Provider adoption fails — clinicians won't input data	Pharmacy-first wedge; templated encounter UI; explicit time-saved measurement; free tier for solo practitioners
NIN coverage / NIMC API reliability gaps	Provisional ID path; SLA negotiation with NIMC; cached identity verification snapshots
Regulatory tightening or NDPA enforcement actions	Day-1 DPO; proactive engagement with NDPC; quarterly compliance review; conservative data residency
Connectivity failures in rural areas	Offline-first design; USSD fallback; partnership with mobile operators for zero-rated data on the patient app
Data breach reputational risk (catastrophic)	Defense-in-depth; field-level encryption for highest-sensitivity fields; insurance; pre-drafted breach communications
HMS vendor opposition (they see us as competition)	Position as a data layer they integrate to, not a replacement; revenue share for adapters; co-marketing
Hospital systems refuse open standards	Begin with smaller and more agile providers; let payer leverage create top-down pull from larger institutions
Funding gap before revenue ramp	Capital-light initial rollout; grant funding (Gates Foundation, USAID, AfDB health programs); phased geography
Patient privacy panic ("government will see my records")	Strong patient-ownership messaging; transparent audit; no government read access without patient consent or court order; community-trusted ambassadors
Clinical liability if data is wrong	Clear ToS placing clinical decisions with the provider; structured data validation; provider attestation on each encounter

18. Open Questions & Decisions Needed

- Final choice of cloud provider for in-country residency (MainOne, MTN, Galaxy Backbone, AWS Lagos, or hybrid).
- Decision on whether to build the patient app as native (Swift/Kotlin) or React Native. Tradeoff: development velocity vs. offline performance.
- Whether to pursue NHIA accreditation as a vendor in year 1 or year 2.
- Pricing elasticity testing with pharmacy and clinic segments — should be done in beta.

- Decision on emergency break-glass policy: is patient SMS sufficient, or should we require post-event provider re-justification?
- Branding and naming for the patient identifier — "MedRecords ID" works internally but needs consumer research.
- Open-sourcing the Nigeria FHIR Implementation Guide (likely yes — it builds ecosystem credibility).

19. Appendix: Glossary

Term	Definition
EHR	Electronic Health Record — institutional clinical record system
PHR	Personal Health Record — patient-owned, patient-controlled record
FHIR	Fast Healthcare Interoperability Resources — HL7's modern interop standard
NIN	National Identification Number — Nigeria's national ID, issued by NIMC
NIMC	National Identity Management Commission — Nigerian agency for NIN
NDPA	Nigeria Data Protection Act 2023
NDPC	Nigeria Data Protection Commission — regulator under NDPA
NHIA	National Health Insurance Authority — Nigeria's national insurance scheme
HMS	Hospital Management System — institutional admin/billing software
MDCN	Medical and Dental Council of Nigeria
PCN	Pharmacists Council of Nigeria
NAFDAC	National Agency for Food and Drug Administration and Control
RBAC / ABAC	Role / Attribute-Based Access Control
Break-glass	Emergency access overriding normal consent, with full audit
LGA	Local Government Area — Nigerian administrative division